

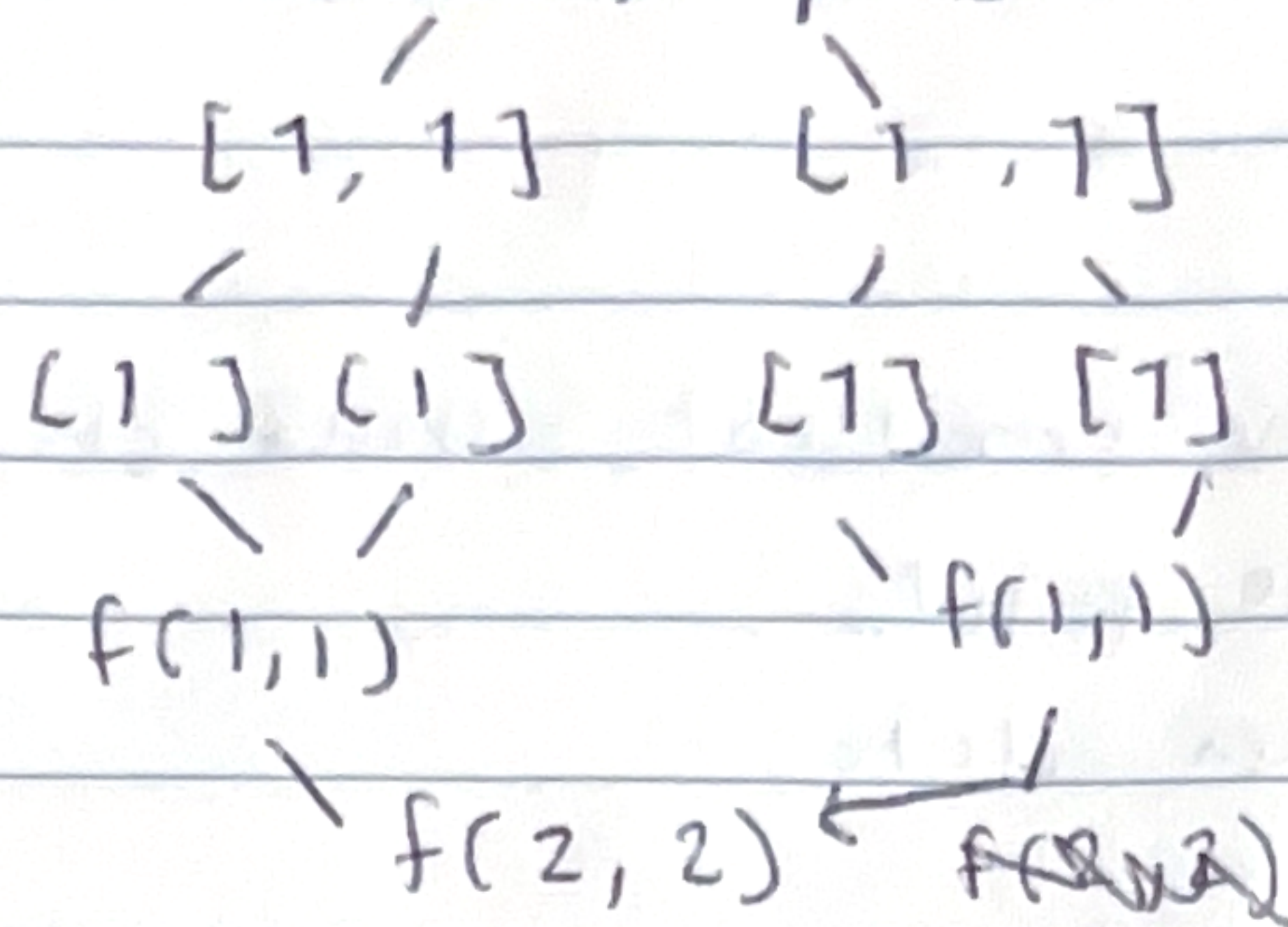
Recitation 4 answers

4. Work and span of word-count-reduce

(parallel reduce)

this function basically adds up everything in a list

example $[1, 1, 1, 1]$



output = 4

basically divide and conquer

- Every single element gets added together exactly once, then for n words items/word count, the work is $O(n)$

- Span: if we had infinite processors, the span would simply be the height of the tree. Since it's being divided by 2, the height is $\log_2 n$. So span is $O(\log n)$

- Also for work, we could say it is root dominated ($n, n/2, n/4, \dots$) because it keeps getting smaller $\rightarrow O(n)$.

5.

the for loops are the issue.

whenever counts gets updated, it has to get the previous value, add 1 to it, and update it. Because of this, if a word goes through one branch and the same word appears again and goes on a separate, they will both get, add 1, and update and without communicating to each other